Example 1: Compute the determinant of

$$A=\begin{bmatrix}1 & 5 & 0\\ 2 & 4 & -1\\ 0 & -2 & 0\end{bmatrix}$$

Solution: $\det A_{11}=\begin{vmatrix}4 & -1\\ -2 & 0\end{vmatrix}=4\cdot 0-(-1)(-2)=-2$

$$\det A_{12}=\begin{vmatrix}2 & -1\\ 0 & 0\end{vmatrix}=2\cdot 0-(-1)\cdot 0=0$$

$$\det A_{13}=\begin{vmatrix}2 & 4\\ 0 & -2\end{vmatrix}=2\cdot(-2)-4\cdot 0=-4$$

By defn, $\det A = 1\cdot\det A_{11}-5\cdot\det A_{12}+0\cdot\det A_{13}$

$= 1\cdot(-2)-5\cdot 0+0\cdot(-4)=-2$

Given $A=[a_{ij}]$, the **(i,j)-cofactor** of A is the number C_{ij} given by

$$C_{ij}=(-1)^{i+j}\det A_{ij} \quad (4).$$

Then $\det A = a_{11}C_{11}+a_{12}C_{12}+\dots\dots+a_{1n}C_{1n}$

This formula is called a **cofactor expansion across the first row** of A.

Theorem 1: The determinant of an $n\times n$ matrix A can be computed by a cofactor expansion across any row or down any column. The expansion across the ith row using the cofactors in (4) is

$$\det A = a_{i1}C_{i1}+a_{i2}C_{i2}+\dots\dots+a_{in}C_{in}$$

The cofactor expansion down the jth column is

$$\det A = a_{1j}C_{1j}+a_{2j}C_{2j}+\dots\dots+a_{nj}C_{nj}$$

The proof is committed.

The plus or minus sign in the (i,j)-cofactor depends on the position of a_{ij} in the matrix, regardless of the sign of a_{ij} itself. The factor $(-1)^{i+j}$ determines the following checkerboard patterns of signs:

$$\begin{bmatrix}+ & - & + & \dots\\ - & + & - & \dots\\ + & - & + & \dots\\ \vdots & & & \end{bmatrix}$$